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Career Highlights

AI Engineer with 3.5+ years of experience building **production-grade AI systems and scalable data platforms**. Specialized in **LLM systems (RAG, fine-tuning, agentic workflows)** and **distributed data engineering**. Proven track record of improving **latency, accuracy, and cost efficiency** through system design, optimization, and robust ML pipelines. Strong focus on **end-to-end deployment, scalability, and reliability** in real-world environments.

AI/ML & Data Engineering Stack

Generative AI & ML: LLMs (OpenAI, LLaMA, Gemini), RAG, Self-RAG, Agentic AI, LoRA/PEFT, Transformers (BERT, LayoutLM), Prompt Engineering, Embeddings, Vector Databases (FAISS), MLflow.

Distributed Systems & Data Engineering: Apache Spark, PySpark, Structured Streaming, Kafka, Delta Lake, ETL/ELT Pipelines, Data Modeling, Data Validation.

Cloud & Infrastructure: Azure Databricks, ADLS Gen2, Azure Data Factory, Synapse Analytics, Snowflake, PostgreSQL.

Backend & MLOps: Python, SQL, FastAPI, Docker, CI/CD (GitHub Actions, Azure DevOps), Async Processing, Caching (Redis).

Professional Experience

Data Engineer (AI Systems & Data Platforms) | Ingenious e-Brain, 06 Jan 2025 – 23 May 2025

- Built **distributed PySpark pipelines** (Azure Databricks) processing 10M+ records/week; reduced runtime by **40%** via partitioning, caching, and optimized joins.
- Designed **real-time streaming architecture (Kafka + Spark)** handling 5K events/min; reduced end-to-end latency from **3–5s < 500ms (85%)**.
- Implemented **Delta Lake (ACID, schema evolution, time travel)** enabling reliable incremental processing and consistent data pipelines.
- Optimized workloads using **predicate pushdown, partition pruning, and memory tuning**, improving throughput by **35–40%**.
- Developed **low-latency pipelines** powering near real-time dashboards (minutes → sub-second freshness).
- Automated ETL with **data validation frameworks**, reducing schema mismatch and null-related issues in production.
- Integrated pipelines with **Snowflake + ADLS** and deployed using CI/CD for reproducible workflows.

Data Analyst (ML-Ready Data Systems) | Provana India, 07 Jan 2021 – 30 Dec 2024

- Built scalable ETL pipelines using PySpark and Snowflake, reducing data processing time from **8 hours to 4.5 hours (40%)** through query optimization and parallelization.
- Designed and implemented data warehouse pipelines, improving query latency for analytics workloads by **30–40%**.
- Developed multi-source data integration frameworks, replacing manual data stitching with automated pipelines.
- Implemented data validation and governance checks, significantly reducing inconsistencies observed in legacy reporting systems.
- Optimized complex SQL queries (joins, indexing strategies), improving execution efficiency across reporting pipelines.
- Delivered ML-ready datasets with consistent schema and feature engineering support for downstream data science use cases.

Key Projects

LLM-Based Personal AI Chatbot (RAG + Self-RAG)

- Built **end-to-end RAG pipeline**: ingestion → chunking → embeddings → retrieval → generation → validation.
- Implemented **Self-RAG validation loop** (grounding + confidence scoring) to reduce hallucinations.
- Improved hallucination rate from **40% → 26% (~35%)**.
- Increased retrieval success rate from **55% → 69% (25%)** via query reformulation and dynamic top-k.
- Enabled **multi-turn conversation memory and context tracking**.
- Deployed modular APIs using **FastAPI + Docker**.

Tech: LangChain, FAISS, Python

LLM Fine-Tuning & Optimization Pipeline

- Fine-tuned LLM using **LoRA (PEFT)** for domain-specific QA tasks.
- Improved accuracy from **62% → 78%** and F1 score from **0.68 → 0.82**.
- Reduced training cost by **70–80%** compared to full fine-tuning.
- Optimized inference latency from **1s → ~600ms** using prompt compression.
- Implemented **caching + async batching**, improving throughput **2–3x**.
- Designed **model routing system**, reducing inference cost by **50–55%**.

Tech: HuggingFace, PEFT, Python, FastAPI

Computer Vision System (Breast Cancer Classification)

- Built **EfficientNet-based model** for benign vs malignant classification with Grad-CAM explainability.
- Improved AUC from **0.75** → **0.88–0.90** using transfer learning and augmentation.
- Addressed class imbalance using **weighted loss functions**.
- Designed full pipeline: **preprocessing** → **training** → **evaluation** → **visualization**.
- Integrated **Grad-CAM** for clinical interpretability.

Tech: PyTorch, OpenCV, EfficientNet

Scalable LLM Serving & RAG Platform (System Design)

- Designed **distributed LLM serving architecture**: API Gateway → Query Router → Retrieval → LLM → Validation → Cache.
- Built **hybrid retrieval system (BM25 + embeddings)** improving relevance for long-tail and ambiguous queries.
- Implemented **FAISS indexing (IVF + HNSW)** to scale retrieval over **1M+ document chunks**.
- Developed **LLM routing system** (small/medium/large models) reducing inference cost by **40–55%** while maintaining accuracy.
- Integrated **multi-level caching (query, embedding, response)** reducing repeated query latency to **<100ms**.
- Optimized inference using **async batching + parallel retrieval**, achieving **<700ms P95 latency**.
- Designed **microservices-based scaling architecture** enabling independent scaling of retrieval and inference layers.
- Implemented **monitoring (latency, token usage, failures)** and fallback mechanisms (re-retrieval, confidence scoring).

Tech: FastAPI, FAISS, Redis, Docker, Python

Certifications: - **Microsoft Certified**: Azure AI Engineer, Data Scientist, Data Engineer, Data Analyst

Education

Master of Technology in AI & ML, Lovely Professional University.	2025 - Present
Postgraduate in Advance Data Science and ML, IIT Madras.	2023 - 2024
Bachelor of Technology in AI & ML, Lovely Professional University.	2016 - 2020